

Foundations of Data Science

Lecture 1: Build, compute, critique, repeat

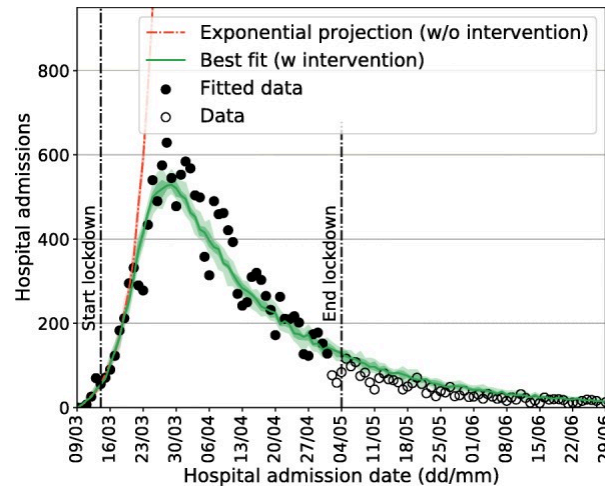
Prof. Gilles Louppe
g.louppe@uliege.be

Three stories

Before any theory, three stories from the last few years.

Each one follows the same four steps: **build** a model of how the data came to be, **compute** what the data say, **critique** the answer, and **repeat**.

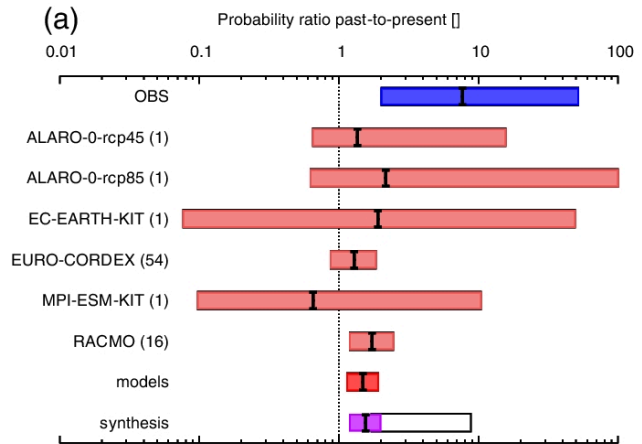
COVID-19 in Belgium



Belgian hospital admissions in 2020 (dots), with and without lockdown.

- **Build:** the Belgian population, its age groups, contacts and mobility.
- **Compute:** the first wave with the lockdown (green), and without it (red).
- **Critique:** after the May 2020 reopening, the second rise never came.
- **Repeat:** behaviour had changed; the models had to include it.

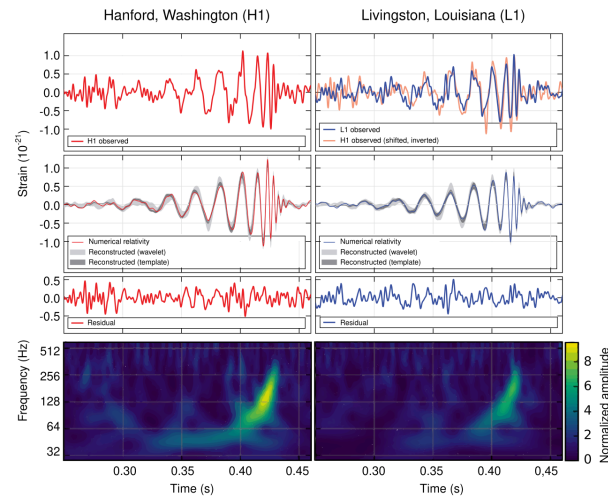
Floods in the Vesdre valley



Pepinster after the floods of July 2021, which killed 38 people in Belgium.

- **Build:** rainfall statistics, and climate models of two possible worlds.
- **Compute:** such rainfall is 1.2 to 9 times more likely today (right).
- **Critique:** keep only the models that reproduce observed rainfall.
- **Repeat:** a rapid study within weeks, a peer-reviewed analysis in 2023.

Gravitational wave detection



First direct detection of gravitational waves, LIGO, 2015.

- **Build:** general relativity predicts the waveform of merging black holes.
- **Compute:** match the data to the waveforms, and infer masses and distance.
- **Critique:** years without a detection, and fake signals to test the analysis.
- **Repeat:** better detectors and noise models, until the signal of 2015.



Box's loop

All models are wrong, but some are useful.

— George Box



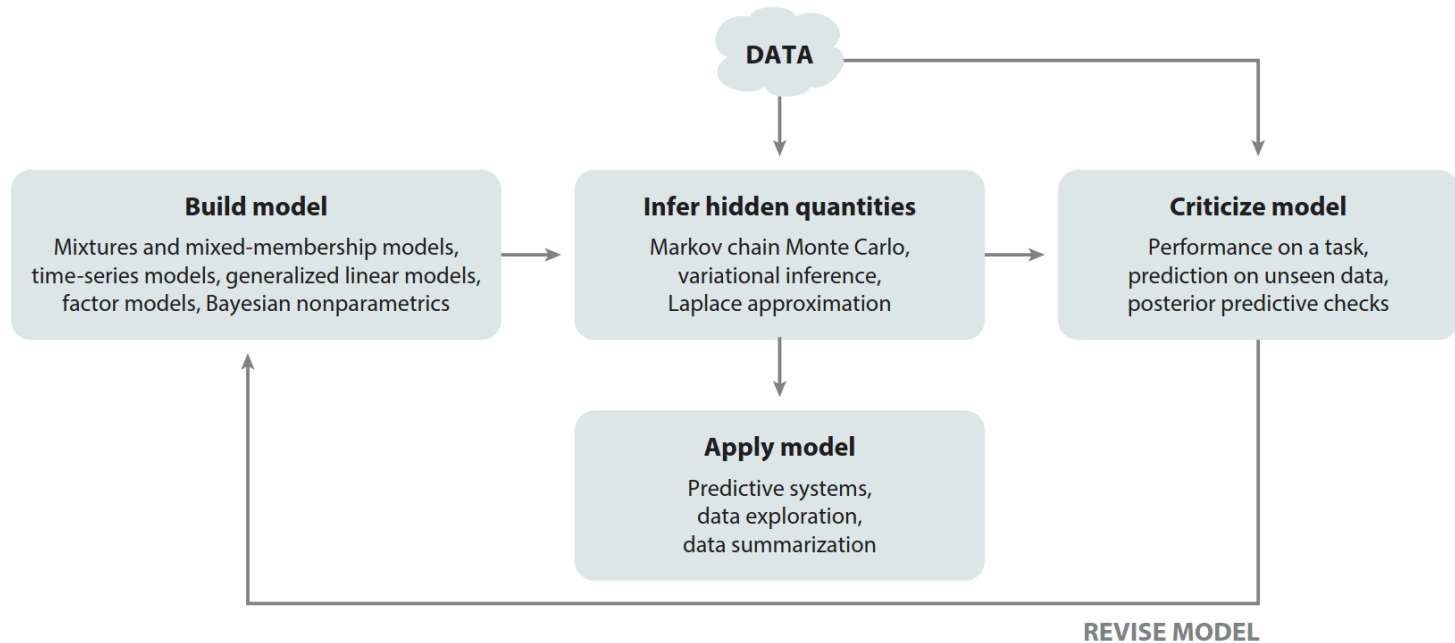
What is data science?

Data science is the discipline of **extracting knowledge** from data through the iterative application of the **scientific method**.

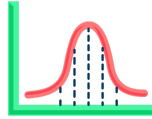
But how?



Box's loop



Scientific inquiry as an iterative process: build, compute, critique, repeat.



Step 1: Build

The first step to understanding a phenomenon is to **build a model** of it, as a simplified representation that captures its essential aspects.

- A model specifies assumptions about the data generating process.
- It encodes domain knowledge and constraints.
- It is formulated within an appropriate mathematical abstraction.
- It defines what you observe and what you do not observe but assume exists.



Step 2: Compute

The next step is to **compute** what the data tells you about the phenomenon of interest under the assumptions of your model.

- Fitting the model to data involves solving an optimization problem.
- Inference runs the model backward, from observed data to the unobserved quantities that could have produced them.
- Prediction is used to answer questions about future or unseen data.



Step 3: Critique

The third step is to **critique** the model and its predictions, to assess whether they are consistent with the data and domain knowledge.

- Compare predictions to observed data.
- Identify model limitations and mismatches.
- Reject the model if it fails to capture key aspects of the data.

Whenever you are fitting a model, we recommend constructing a scenario where it will fail.

— Gelman et al.

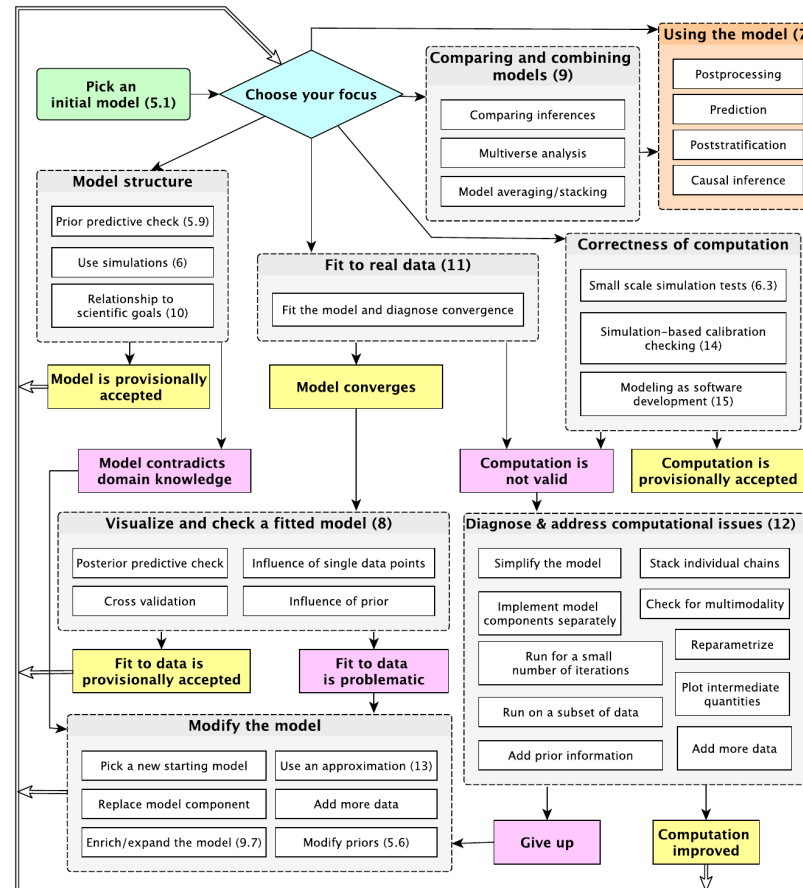


Step 4: Repeat

What you learn from the critique step informs how to **repeat** the process.

- Add complexity to the model to address its shortcomings.
- Simplify the model to improve interpretability.
- Change the model to explore alternative hypotheses.

The loop, in practice



Bayesian workflow: the steps and paths an analysis may go through.

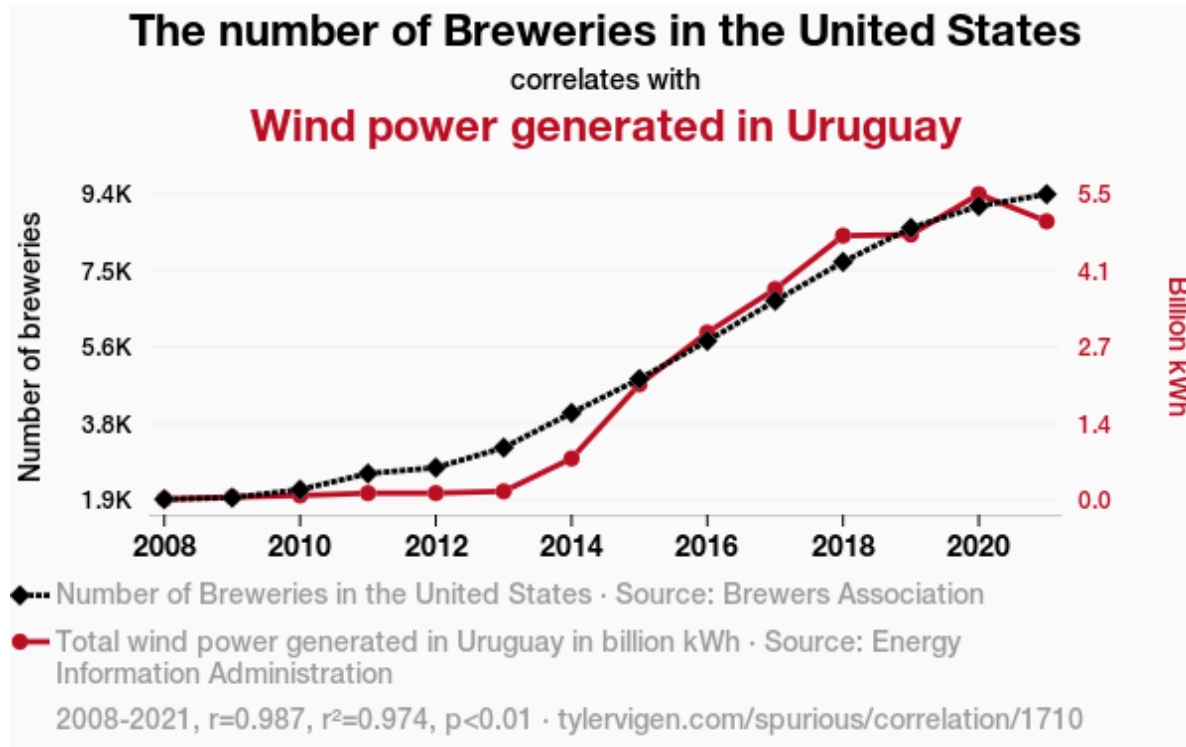
Why this approach matters?

The Fourth paradigm (Hey et al, 2009) of science emphasizes the importance of data-intensive scientific discovery. However, data alone is not enough.

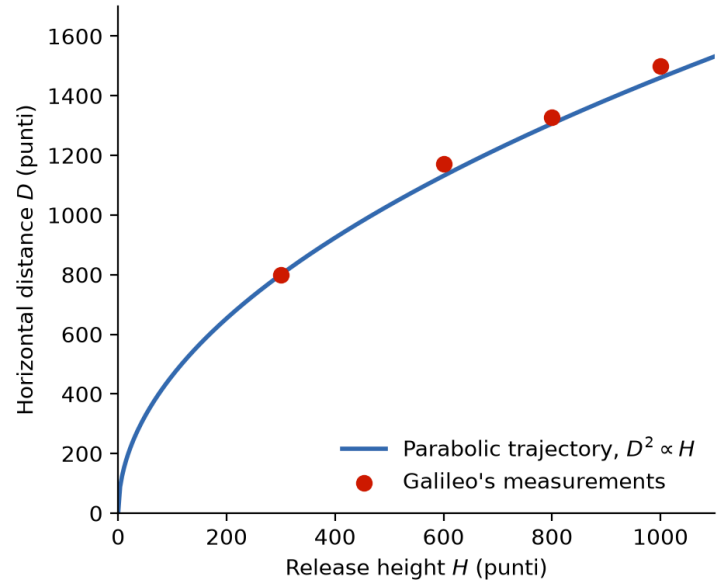
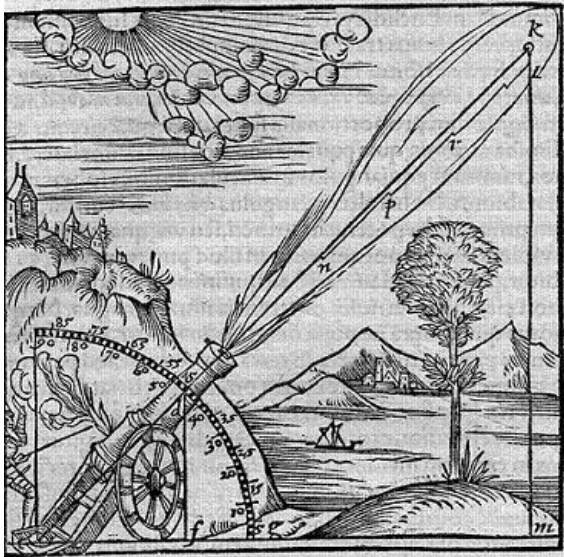
- Data without theory leads to spurious correlations.
- Theory without data leads to ungrounded speculation.
- Together, they enable **robust scientific inquiry**.

It is frequently assumed that big data can alleviate the need for careful modeling. We do not believe this is the case.

— Gelman et al.



Spurious correlation can easily mislead data analysis.



Theory without data: the trajectory implied by Aristotelian physics (left), against Galileo's measurements, consistent with a parabola (right).

The scientific method, as embodied in Box's loop, provides a principled framework for data analysis. It contrasts with **ad-hoc data analysis practices that often lead to unreliable results**.

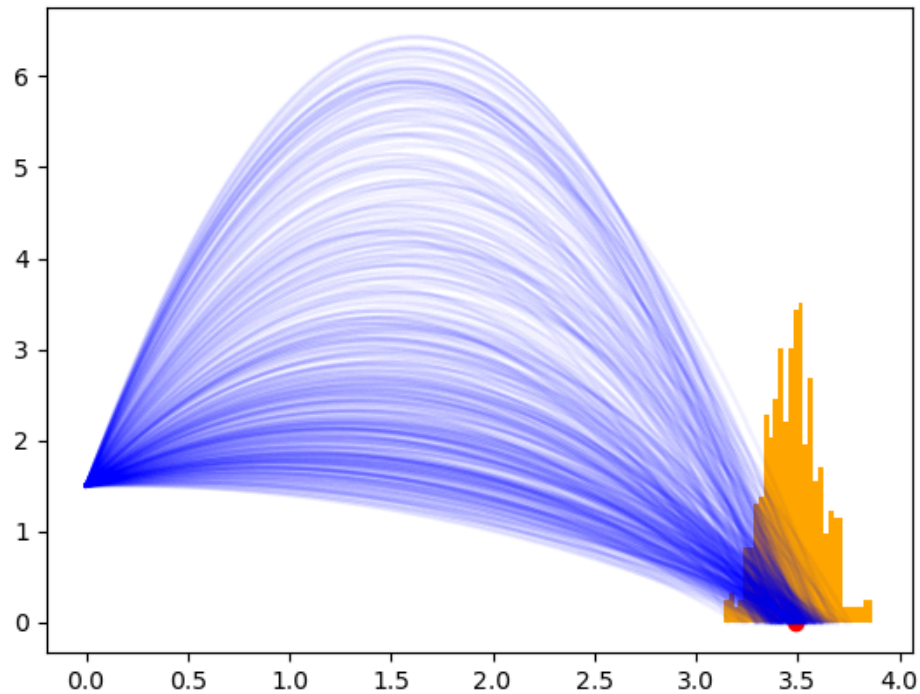
- Throw algorithms at data and see what sticks.
- Focus only on prediction accuracy.
- Treat models as black boxes.
- Stop at the first somewhat satisfactory result.

Box's loop encourages

- transparent reasoning about assumptions,
- understanding mechanisms behind data, not just correlations,
- honest assessment of model limitations,
- continuous improvement and learning.

Today's example: Projectile motion

A ball is thrown and lands at some measured distance x . What can we infer about the initial velocity v and angle α of the throw?



Let's code!

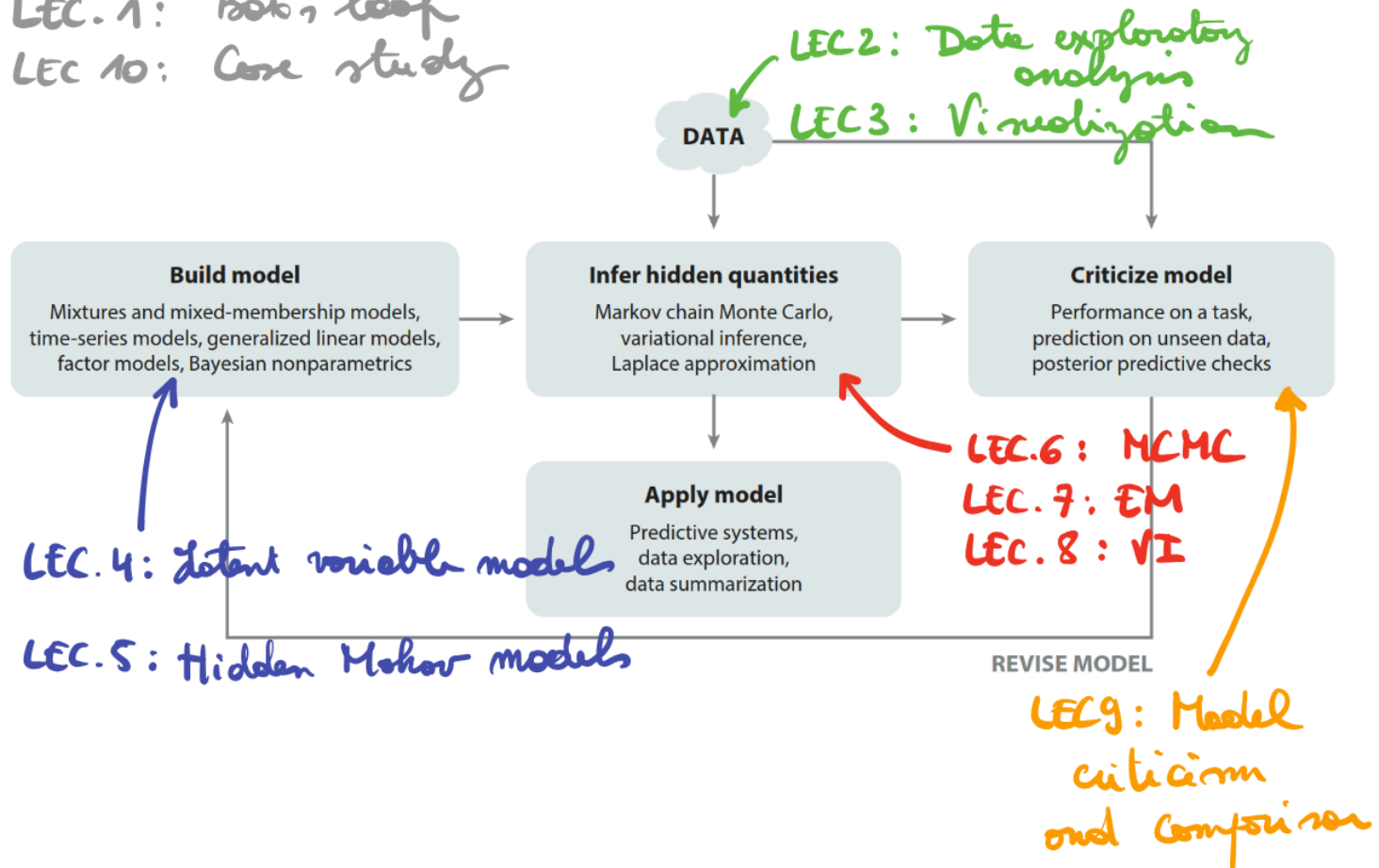
- Open a Jupyter notebook and follow along (or check `nb01.ipynb` after the lecture).
- Implement Box's loop step by step:
 1. Build a simple physical model of projectile motion.
 2. Compute estimates of $\mathbf{v}$ and α from data.
 3. Critique the model fit and predictions.
 4. Repeat by refining the model to account for more realistic factors.

DATS0001

Outline

- Lecture 1: Build, compute, critique, repeat
- Lecture 2: Data and exploratory analysis
- Lecture 3: Visualization
- Lecture 4: Latent variable models
- Lecture 5: State-space models
- Lecture 6: Markov Chain Monte Carlo
- Lecture 7: Model criticism and validation
- Lecture 8: Expectation-Maximization
- Lecture 9: Variational Inference
- Lecture 10: Simulation-based inference
- Lecture 11: Case study

LEC. 1: Box's loop
LEC 10: Core study



My mission

Teach you how to think like a scientist in the age of data.

By the end of this course, you will be able to explore, model, and reason about data in a principled way, but also to communicate your findings effectively.

The end.